according to the order of time.

“According to the order of time” (κατὰ τὴν τοῦ χρόνου τάξιν). From one of the crucial, initial moments of natural science there remains nothing but these obscure, arcanely resonant words, this appeal to the “order of time.”

Astronomy and physics have since developed by following this seminal lead given by Anaximander: by understanding how phenomena occur *according to the order of time*. In antiquity, astronomy described the movements of stars *in time*. The equations of physics describe how things change *in time*. From the equations of Newton, which establish the foundations of mechanics, to those of Maxwell for electromagnetic phenomena; from Schrödinger’s equation describing how quantum phenomena evolve, to those of quantum field theory for the dynamics of subatomic particles: the whole of our physics, and science in general, is about how things develop “according to the order of time.”

It has long been the convention to indicate this time in equations with the letter *t* (the word for “time” begins with *t* in Italian, French, and Spanish, but not in German, Arabic, Russian, or Mandarin). What does this *t* stand for? It stands for the number measured by a clock. The equations tell us how things change as the time measured by a clock passes.

But if different clocks mark different times, as we have seen above, what does *t* indicate? When the two friends meet up again after one has lived in the mountains and the other at sea level, the watches on their wrists will show different times. Which of the two is *t*? In a physics laboratory, a clock on a table and another on the ground run at different speeds. Which of the two tells the time? How do we describe the difference between them? Should we say that the clock on the ground has slowed relative to the real time recorded on the table? Or that the clock on the table runs faster than the real time measured on the ground?

The question is meaningless. We might just as well ask what is *most* real—the value of sterling in dollars or the value of dollars in sterling. There is no “truer” value; they are two currencies that have value *relative to each other*. There is no “truer” time; there are two times and they change *relative to each other*. Neither is truer than the other.

But there are not just *two* times. Times are legion: a different one for every point in space. There is not one single time; there is a vast multitude of them.

The time indicated by a particular clock measuring a particular phenomenon is called “proper time” in physics. Every clock has its proper time. Every phenomenon that occurs has its proper time, its own rhythm.